

PURESOF

Professional Training Program

Training Report

Data Analysis Track

Prepared by

Omar Sharaby

Comprehensive Technical & Capstone Project Documentation

Table of Contents

1.	Advanced Excel & Business Analytics	Page 3
2.	Python for Data Analysis & Exploratory Data Analysis (EDA)	Page 4
3.	Applied Mathematics & Statistical Foundations	Page 5
4.	Relational Database Design & Advanced SQL Querying	Page 6
5.	Advanced SQL for Analytics, Window Functions & Optimization	Page 7
6.	Data Warehousing, Dimensional Modeling & Power Query ETL	Page 8
7.	Power BI Architecture, DAX Modeling & Business Intelligence	Page 9
8.	Enterprise Storage Modes: DirectQuery vs. Import Benchmark	Page 10
9.	Enterprise Security Governance: Dynamic Row-Level Security (RLS)	Page 11
10.	Capstone Project: Enterprise Operational Intelligence Platform (Architecture & Model)	Page 12
11.	Capstone Project: Diagnostic Root Cause Analysis (RCA Findings)	Page 13
12.	Capstone Project: Dynamic RLS Implementation & Dashboard Layout	Page 14
13.	Capstone Project: Benchmarking, Technical Scale & Evaluation	Page 15
14.	Technology Stack & Executive Summary	Page 16

Puresoft Data Analysis Track Objectives & Learning Outcomes

This technical training report documents the comprehensive competencies acquired across the four core pillars of the **Puresoft Data Analysis Track**: (1) Cleaning and preparing raw multi-source business data, (2) Performing exploratory data analysis (EDA) using Python (Pandas/NumPy), (3) Formulating complex SQL queries to extract, transform, and manipulate relational data, and (4) Engineering high-impact interactive dashboards in Power BI. The program culminated in a production-grade Capstone Intelligence solution resolving systemic enterprise operational and pricing bottlenecks.

Advanced Excel & Business Analytics

- Advanced Lookup & Reference Architecture

Mastering dynamic data retrieval using INDEX-MATCH two-way lookups, modern XLOOKUP with exact match, wildcard matching, and reverse searching, eliminating legacy limitations of VLOOKUP regarding column indexing and directional constraints.

- Complex Logical & Text Processing Formulas

Structuring nested multi-criteria logical evaluations using IFS, AND, OR, and boolean masking. Extracting and transforming irregular strings with TEXTSPLIT, LET, LAMBDA, LEFT, MID, FIND, and regex patterns to cleanse unstructured customer and operational records.

- PivotTables, Multi-Dimensional Summaries & Slicers

Constructing enterprise aggregation summaries, calculating running totals, % of parent row/column, and custom calculated fields. Implementing interactive multi-table dashboards synchronized across timelines and cross-filtering slicers.

- What-If Decision Modeling & Scenario Analysis

Evaluating financial risk and pricing sensitivity through Goal Seek, Scenario Manager, and Two-Variable Data Tables. Modeling breakeven analysis under shifting unit cost structures and volume discount fluctuations.

- Data Validation, Auditing & Error Handling

Securing corporate workbooks against invalid data entry using custom validation rules, dependent drop-down lists (Indirect formula), formula auditing via trace precedents/dependents, and encapsulating risk points using IFERROR and ISBLANK.

- Power Query for Excel (In-Sheet ETL Automation)

Automating repetitive ingestions from folders, CSVs, and web endpoints. Executing multi-step schema transformations, unpivoting cross-tabulated financial reports into normalized tabular data, and building reusable M transformations prior to analytical modeling.

Analytical Technique	Excel Implementation	Business Application
Dynamic Matrix Lookup	INDEX(Range, MATCH(), MATCH())	Tiered freight pricing lookup based on weight and postal zone.
Modern Fast Search	XLOOKUP(Key, LookupCol, ReturnCol, "N/A")	Instant SKU price retrieval resilient to column insertions.
Dynamic Normalization	Power Query Unpivot Columns	Restructuring 12-month budget columns into normalized Date/Amount rows.
Sensitivity Simulation	Data Tables (Row & Column inputs)	Visualizing gross margin variance across pricing adjustments and inflation.

Python for Data Analysis & Exploratory Data Analysis

- **High-Performance Numerical Computing (NumPy)**

Leveraging n-dimensional array architectures (`ndarray`) for memory-efficient vectorized computations, multi-dimensional slicing, mathematical broadcasting, and boolean index filtering without the performance penalties of standard Python loops.

- **Structured Tabular Analytics (Pandas DataFrames & Series)**

Loading heterogeneous datasets from CSV, Excel, JSON, and relational SQL engines. Slicing with `.loc[]` and `.iloc[]`, hierarchical multi-indexing, data reshaping via `.pivot_table()` and `.melt()`, and executing high-speed aggregations with `.groupby()`.

- **Rigorous Data Cleansing & Quality Assurance Pipelines**

Detecting and imputing missing records (mean/median/KNN imputation vs. targeted drop), eliminating duplicates, correcting mismatched data types (categorical encoding, timestamp parsing), and sanitizing string columns via vectorized regex (`.str.replace()`).

- **Exploratory Data Analysis (EDA) & Feature Inspection**

Systematic dataset profiling to uncover distributions, anomalies, and covariance. Computing summary statistics (five-number summary, variance, IQR), identifying outliers through Tukey's boxplot fence rule, and formulating empirical hypotheses.

- **Visual Analytics & Statistical Storytelling (Matplotlib & Seaborn)**

Building expressive statistical visualizations: univariate histograms and KDE plots for skewness inspection, bivariate scatter plots with regression trends, correlation heatmaps with Pearson coefficients, and categorical faceted violin/box plots.

```
# Production Data Cleaning & Grouped EDA Pipeline Example
import pandas as pd
import numpy as np

def clean_and_profile_sales(df: pd.DataFrame) -> pd.DataFrame:
    # 1. Type casting and timestamp parsing
    df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
    df['UnitPrice'] = pd.to_numeric(df['UnitPrice'], errors='coerce')

    # 2. Impute missing values & remove cancelled transactions
    df = df.dropna(subset=['CustomerID', 'UnitPrice']).copy()
    df = df[df['Quantity'] > 0]

    # 3. Vectorized feature engineering
    df['LineTotal'] = df['Quantity'] * df['UnitPrice']
    df['GrossMarginPct'] = (df['LineProfit'] / df['LineTotal']).clip(-1.0, 1.0)

    # 4. Outlier isolation via Interquartile Range (IQR)
    q1, q3 = df['LineTotal'].quantile([0.25, 0.75])
    iqr = q3 - q1
    df['IsOutlier'] = (df['LineTotal'] < (q1 - 1.5 * iqr)) | (df['LineTotal'] > (q3 + 1.5 * iqr))
    return df
```

Applied Mathematics & Statistical Foundations

- Descriptive Statistics & Data Dispersion Metrics**

Quantifying central tendency (arithmetic mean, median, mode) and understanding the vulnerability of the mean to extreme values. Measuring statistical dispersion via sample variance, standard deviation, and Interquartile Range (IQR) to assess enterprise stability.

- Probability Distributions & Central Limit Theorem (CLT)**

Analyzing theoretical distributions including Normal (Gaussian), Binomial, and Poisson models. Leveraging the Central Limit Theorem to guarantee that sample mean distributions approach normality, enabling parametric estimation on large operational populations.

- Inferential Statistics, Confidence Intervals & Hypothesis Testing**

Formulating null (H_0) and alternative (H_1) hypotheses for business decision-making. Computing z-scores, Student's t-tests (two-sample independent and paired), ANOVA for multi-group mean comparisons, and Chi-Square (χ^2) tests for independence, evaluating significance against p-value thresholds ($\alpha = 0.05$).

- Correlation vs. Causation & Regression Modeling**

Distinguishing between Pearson linear correlation (r) and Spearman monotonic rank correlation. Formulating Ordinary Least Squares (OLS) simple and multiple linear regressions to quantify elasticity, interpreting R^2 variance determination, and checking homoscedasticity.

Statistical Method	Formula / Test Statistic	Business Intelligence Scenario
Standardized Z-Score	$Z = \frac{X - \mu}{\sigma}$	Detecting anomalous invoice quantities and billing spikes.
Two-Sample t-Test	$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_p^2(1/n_1 + 1/n_2)}}$	Validating whether a discount campaign significantly increased basket size.
Chi-Square Test (χ^2)	$\chi^2 = \sum \frac{(O - E)^2}{E}$	Testing correlation between regional territory and order cancellation rates.
Coefficient of Determination	$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$	Measuring what percentage of warehouse delivery delays is driven by carrier lag.

Relational Database Design & Advanced SQL

- **Relational Architecture & Normalization Principles**

Designing normalized databases (1NF atomic values, 2NF no partial dependencies, 3NF no transitive dependencies) to guarantee data integrity, eliminate update anomalies, and enforce primary keys, foreign keys, unique constraints, and check validation rules.

- **Data Definition (DDL) & Data Manipulation (DML)**

Executing robust schema creation scripts (`CREATE TABLE` , `ALTER` , `DROP`) with optimized datatypes (`INT` , `DECIMAL(18,2)` , `NVARCHAR`). Managing ACID-compliant transactional state changes via `INSERT` , `UPDATE` , `DELETE` , and `MERGE` within transaction blocks (`BEGIN TRAN` / `COMMIT`).

- **Relational Joins & Set Algebra**

Mastering relational combinations using `INNER JOIN` , `LEFT OUTER JOIN` , `RIGHT OUTER JOIN` , `FULL OUTER JOIN` , and Cartesian `CROSS JOIN` . Combining disjoint record sets through `UNION` , `UNION ALL` , `INTERSECT` , and `EXCEPT` .

- **Multi-Level Aggregation & Dimensional Grouping**

Aggregating transaction volumes with `COUNT` , `SUM` , `AVG` , `MIN` , and `MAX` . Enforcing post-aggregation filtering with `HAVING` clauses and generating hierarchical multidimensional subtotals using `GROUP BY ROLLUP` and `CUBE` .

```
-- Multi-Table Enterprise Sales Ingestion with Customer Classification
SELECT
    c.CustomerID,
    c.CustomerName,
    cc.CustomerCategoryName,
    COUNT(DISTINCT i.InvoiceID) AS TotalInvoicesPlaced,
    SUM(il.Quantity) AS TotalUnitsPurchased,
    CAST(SUM(il.Quantity * il.UnitPrice) AS DECIMAL(18,2)) AS TotalGrossRevenue,
    CAST(SUM(il.LineProfit) AS DECIMAL(18,2)) AS TotalOperatingProfit,
    CAST(SUM(il.LineProfit) / NULLIF(SUM(il.Quantity * il.UnitPrice), 0) * 100 AS DECIMAL(5,2)) AS Margin
FROM Sales.Customers c
INNER JOIN Sales.Invoices i ON c.CustomerID = i.CustomerID
INNER JOIN Sales.InvoiceLines il ON i.InvoiceID = il.InvoiceID
LEFT JOIN Sales.CustomerCategories cc ON c.CustomerCategoryID = cc.CustomerCategoryID
GROUP BY
    c.CustomerID,
    c.CustomerName,
    cc.CustomerCategoryName
HAVING SUM(il.Quantity * il.UnitPrice) > 50000
ORDER BY TotalGrossRevenue DESC;
```

Advanced SQL for Analytics & Performance Tuning

- **Analytical SQL Window Functions**

Computing rankings, partitions, and moving metrics without costly self-joins: `ROW_NUMBER()` for deduplication, `RANK()` and `DENSE_RANK()` for competitive positioning, and `NTILE()` for cohort segmentation.

- **Time-Series Offsetting & Moving Frames**

Accessing inter-row states with `LAG()` and `LEAD()` to compute period-over-period sales growth, velocity shifts, and churn timing. Specifying dynamic sliding window aggregations via `ROWS BETWEEN N PRECEDING AND CURRENT ROW`.

- **Common Table Expressions (CTEs) & Recursive Logic**

Modularizing multi-phase diagnostic transformations using readable CTEs (`WITH Clause`), simplifying complex nested subqueries, and writing recursive CTEs to traverse organizational hierarchies and multi-tier bill-of-materials structures.

- **Execution Plan Profiling & Index Optimization**

Interpreting graphical and XML execution plans to detect expensive table scans, hash matches, and key lookups. Designing targeted clustered, non-clustered, and covering indexes with `INCLUDE` columns to accelerate analytical read operations.

- **Enterprise Reporting Schema Decoupling**

Creating a dedicated database schema (e.g. `BI`) isolating reporting consumers from internal OLTP application tables. Providing hardened, pre-joined views that eliminate operational table locking and prevent ad-hoc query contention.

```
-- Window Functions: MoM Sales Growth and Top Customer Ranking per Category
WITH MonthlyCustomerSales AS (
    SELECT
        c.CustomerCategoryID,
        c.CustomerID,
        c.CustomerName,
        DATEFROMPARTS(YEAR(i.InvoiceDate), MONTH(i.InvoiceDate), 1) AS SalesMonth,
        SUM(il.LineProfit) AS MonthlyProfit
    FROM Sales.Invoices i
    JOIN Sales.InvoiceLines il ON i.InvoiceID = il.InvoiceID
    JOIN Sales.Customers c ON i.CustomerID = c.CustomerID
    GROUP BY c.CustomerCategoryID, c.CustomerID, c.CustomerName,
             DATEFROMPARTS(YEAR(i.InvoiceDate), MONTH(i.InvoiceDate), 1)
)
SELECT
    CustomerCategoryID, CustomerName, SalesMonth, MonthlyProfit,
    LAG(MonthlyProfit, 1) OVER (PARTITION BY CustomerID ORDER BY SalesMonth) AS PriorMonthProfit,
    DENSE_RANK() OVER (PARTITION BY CustomerCategoryID, SalesMonth ORDER BY MonthlyProfit DESC) AS Cate
FROM MonthlyCustomerSales;
```

Data Warehousing, Dimensional Modeling & Power Query

- **Kimball Dimensional Modeling Methodology**

Translating transactional OLTP schemas into analytical OLAP structures. Segregating business metrics into numeric **Fact Tables** and descriptive business contexts into **Dimension Tables** to optimize analytical query throughput.

- **Star Schema vs. Snowflake Schema Architecture**

Designing clean Star Schemas where fact tables connect directly to single-level denormalized dimensions via single-column integer surrogate keys, minimizing table joins, eliminating ambiguous filter paths, and maximizing VertiPaq compression.

- **Fact Granularity & Slowly Changing Dimensions (SCD)**

Establishing strict grain definitions (e.g. one row per invoice line item). Implementing strategies for changing customer/product states over time: SCD Type 1 (overwrite), SCD Type 2 (historical versioning with effective/expiration timestamps), and SCD Type 3 (current vs. previous column).

- **Power Query (M Language) Advanced Transformation**

Developing repeatable extract, transform, and load (ETL) pipelines using the functional M formula language: data type enforcement, null handling, column merging, conditional column derivation, and schema stabilization across refreshes.

- **Query Folding Architecture**

Ensuring Power Query steps fold back into native T-SQL queries executed directly inside the SQL Server engine. Preventing local client-side memory bottlenecks by forcing filtering, sorting, and joins to execute at the database tier.

```
// Power Query M Script: Importing Star Schema Dimension with Type Safety
let
    Source = Sql.Database("localhost", "WideWorldImporters", [CreateNavigationProperties=false]),
    DimCustomer_View = Source{[Schema="BI", Item="vw_DimCustomer"]}[Data],
    CleanedTypes = Table.TransformColumnTypes(DimCustomer_View, {
        {"CustomerID", Int64.Type},
        {"CustomerName", type text},
        {"CustomerCategoryName", type text},
        {"SalesTerritory", type text},
        {"CreditLimit", Currency.Type},
        {"IsOnCreditHold", type logical},
        {"PaymentDays", Int64.Type}
    }),
    FilteredActive = Table.SelectRows(CleanedTypes, each [CustomerID] <> null)
in
    FilteredActive
```

Power BI Architecture, DAX & Analytical Modeling

- **Power BI End-to-End Enterprise Architecture**

Designing high-governance business intelligence models: data source ingestion, semantic model relationships, DAX measure engineering, report canvas design, and cloud distribution via Power BI Service workspaces and scheduled gateway refreshes.

- **Calculated Columns vs. Explicit DAX Measures**

Enforcing strict BI best practices: eliminating RAM-heavy calculated columns in favor of explicit DAX measures. Computing aggregations dynamically on demand during visual query evaluation rather than inflating data model storage size.

- **DAX Evaluation Contexts (Filter & Row Context)**

Mastering the fundamental mechanics of DAX: initial visual Filter Contexts, iterative Row Contexts generated by iterator functions (`SUMX` , `AVERAGEX`), and Context Transition triggered when invoking explicit measures inside row iterators.

- **Context Manipulation with CALCULATE**

Overriding, modifying, and restoring filter contexts using `CALCULATE()` . Applying `ALL()` to compute grand total percentages, `ALLEXCEPT()` for subtotal preservation, and `KEEPFILTERS()` for non-destructive filter intersections.

- **Time Intelligence Modeling**

Constructing dedicated calendar tables marked as official Date Tables. Calculating standard financial shifts: Year-to-Date (`TOTALYTD`), Period-over-Period (`DATEADD`), and Prior Year Comparisons (`SAMEPERIODYEAR`) for corporate growth benchmarking.

```
// Core Financial & Time Intelligence DAX Measures
Total Revenue =
SUM(vw_FactSales[TotalAmountIncludingTax])

Total Operating Profit =
SUM(vw_FactSales[LineProfit])

Gross Margin % =
DIVIDE([Total Operating Profit], [Total Revenue], 0)

Revenue Prior Year =
CALCULATE([Total Revenue], SAMEPERIODYEAR(vw_DimDate[DateKey]))

Sales YoY Growth % =
VAR Prev = [Revenue Prior Year]
RETURN DIVIDE([Total Revenue] - Prev, Prev, 0)
```

Enterprise Storage Modes: DirectQuery vs. Import

- Storage Engine Architecture: VertiPaq vs. Relational Pushdown

Benchmarking the two fundamental Power BI storage modes: **Import Mode** loads compressed columnar dictionaries into memory using the VertiPaq engine, whereas **DirectQuery Mode** leaves all data inside SQL Server, converting every user visual interaction into live T-SQL queries.

Architectural Dimension	Import Mode (VertiPaq Engine)	DirectQuery Mode (SQL Server Pushdown)
Storage Engine	In-Memory Columnar Database (VertiPaq)	External Relational Engine (SQL Server)
Data Freshness	Scheduled Refresh (Snapshot)	Real-Time / Live on every filter change
Query Execution	Ultra-fast RAM dictionary evaluations	Formula Engine translates DAX into T-SQL
DAX Capabilities	100% full DAX and Time Intelligence	Restricted DAX; no calculated tables
Database Workload	Zero workload between scheduled refreshes	Continuous concurrent load for every active user

- Empirical Performance Analyzer Benchmark

Stress-testing a complex Matrix visual (Customer Category > Stock Item Name, grouping 228,265 invoice rows across 70,510 orders) using the native Power BI **Performance Analyzer**:

Performance Phase	Import Mode (VertiPaq)	DirectQuery Mode (SQL Server)
DAX Query Time	12 ms – 28 ms	15 ms – 35 ms
DirectQuery (SQL Server Wait)	0 ms (In-Memory)	180 ms – 450 ms (Network & Engine Wait)
Visual Display Render Time	18 ms	22 ms
Total Visual Load Time	~35 ms – 50 ms	~220 ms – 510 ms
Memory Footprint	~14 MB compressed RAM	~2 MB (Metadata only)

- DirectQuery T-SQL Pushdown Inspection

Analyzing the SQL batch emitted by Power BI when evaluating aggregated sales metrics. Demonstrating how Power BI wraps dimensions and fact tables into nested subqueries with TOP (1000001) constraints.

Enterprise Security: Dynamic Row-Level Security

- Principle of Least Privilege & Multi-Tenant BI Governance

Eliminating redundant reports by maintaining a single master dashboard (.pbix). Dynamic Row-Level Security (RLS) dynamically filters underlying tables based on the authenticated user's credentials, ensuring managers strictly view their authorized data.

- Identity Functions: USERPRINCIPALNAME() vs. USERNAME()

Implementing dual-compatibility DAX security rules: retrieving the cloud Microsoft Entra ID UPN (e.g. `user@wideworldimporters.com`) when hosted on Power BI Service, and gracefully falling back to domain workstation credentials (`DOMAIN\User`) in Power BI Desktop.

- Security Bridge Table & Filter Propagation Architecture

Decoupling security rules from dimension tables via a dedicated bridge table (`vw_SecurityUserTerritory`). Filtering the customer dimension automatically filters all associated fact tables (Sales, Backorders, Receivables) through standard 1-to-many relationship propagation.

- Global Executive Wildcard Bypass (*)

Engineering an executive security escape hatch: if a user is assigned the asterisk wildcard (`*`), the DAX rule bypasses territory constraints entirely, granting enterprise-wide visibility to VPs and C-level executives.

- Multi-Persona Testing & Verification Matrix

User Account (UPN / Email)	Role Title	Assigned Territory	Governance Outcome
<code>kaylaw@wideworldimporters.com</code>	Regional Sales Mgr	Southeast	Views FL, GA, NC, SC, TN, AL customers only.
<code>hudson@wideworldimporters.com</code>	Regional Sales Mgr	Far West	Views CA, NV, OR, WA, AK, HI customers only.
<code>lilyc@wideworldimporters.com</code>	Senior Sales Mgr	Southwest & Southeast	Multi-territory filter enabled (TX, AZ, NM, OK + SE).
<code>jackp@wideworldimporters.com</code>	VP of Global Sales	All Territories (*)	Wildcard bypasses filter; views complete enterprise.
<code>omar\omar sharaby</code>	Current Developer	Southeast / Custom	Automatically matches workstation Windows login.

CAPSTONE PROJECT CASE STUDY

Enterprise Operational Intelligence Platform

General Idea & Executive Problem Statement

The Capstone Project represents an enterprise-scale, end-to-end data analytics and operational intelligence system deployed against the Microsoft **WideWorldImporters** wholesale database. The objective was to uncover hidden systemic revenue leaks, supply chain fulfillment failures, pricing misalignment, and overdue debt exposure across over 228,000 transaction line items, translating forensic findings into an interactive, secured Power BI dashboard.

Project Item	Technical Specification & Architecture
Project Name	Enterprise Operational Intelligence & Supply Chain Analytics Platform
Database Engine	Microsoft SQL Server 2022 / Azure SQL Database (Database: WideWorldImporters)
Reporting Layer	Dedicated BI Schema (8 Star-Schema Views separating analytical queries from OLTP)
Storage Architectures	Hybrid Evaluation: VertiPaq In-Memory (Import) & Live Relational (DirectQuery)
Security Governance	Dynamic Row-Level Security (RLS) via USERPRINCIPALNAME() and Territory Bridge Table
Analytics Tools	Power BI Desktop, DAX Studio, Tabular Editor, SQL Server Management Studio (SSMS), Python

Dimensional Star Schema & Relational Model Architecture

The data model is engineered around a centralized Star Schema with 4 specialized fact tables joined to 3 conforming dimensions and 1 hidden dynamic security bridge:

View Name	Table Classification	Analytical Purpose & Granularity
BI.vw_FactSales	Core Fact Table	Invoiced line items with quantity, unit price, tax, and operating profit.
BI.vw_FactUnfulfilledOrders	Diagnostic Fact	Unpicked backorders, lost potential revenue, and supply chain stockout bottlenecks.
BI.vw_FactOverdueReceivables	Financial Fact	Aging outstanding customer invoices, overdue days, and payment terms risk.
BI.vw_FactInventoryHoldings	Inventory Fact	Current warehouse stock-on-hand, reorder thresholds, and target levels.
BI.vw_DimCustomer	Conformed Dimension	Customer demographics, buying groups, categories, credit limits, and sales territories.
BI.vw_DimStockItem	Conformed Dimension	SKU attributes, package types, colors, outer packaging quantities, and list pricing.
BI.vw_DimDate	Calendar Dimension	Comprehensive date attributes supporting standard time intelligence functions.
BI.vw_SecurityUserTerritory	Security Bridge	Maps user UPNs / domain logins to authorized sales territories with wildcard access.

CAPSTONE PROJECT CASE STUDY

Diagnostic Root Cause Analysis (RCA Findings)

Through forensic T-SQL querying and statistical inspection of the WideWorldImporters database, four severe enterprise anomalies were uncovered, quantified, and resolved:

Anomaly Area	Quantified Impact	Affected Volume	Underlying Root Cause
1. Negative Margin Sales	-\$320,493.55 Loss	4,626 Invoice Lines	Packaging math flaw (unit cost exceeds list price) and 10-pack billing bugs.
2. Severe Stockouts	\$5,372,935.20 Lost	3,085 Unpicked Orders	Inadequate safety stock and static reorder levels causing complete inventory depletion.
3. Fulfillment SLA Breaches	6.3% Breach Rate	4,470 Late Shipments	Average 3-day delivery lag concentrated within specific regional freight carriers.
4. Overdue Receivables	\$267,011.44 Debt	84 Open Invoices	57.5% debt concentration in 2 enterprise accounts with zero credit hold enforcement.

Detailed Finding 1: Structural Margin Inversion (-\$320.5K Loss)

Forensic SQL revealed 4,626 sales lines executed at negative operating profit. The primary culprit was Halloween zombie mask (Light Brown) (Sizes S, M, L, XL), accounting for -\$274,500 in losses. Suppliers charged \$228.00 per outer box of 12 (true cost \$19.00/unit), while selling price was misconfigured at \$18.00/unit, guaranteeing a -\$1.00 loss on every unit sold. Additionally, 10-pack USB drives (\$240 value) were repeatedly billed at single-unit prices (\$36), leaking \$15,163 across 43 invoices.

Detailed Finding 2: \$5.37M In Unfulfilled Demand Due to Stockouts

3,085 orders between 2013 and 2016 were never picked (PickingCompletedWhen IS NULL), leaving 360,276 units unsupplied. High-velocity items suffered catastrophic depletion: 67,296 units of packaging dispatch tape were demanded while warehouse stock was just 50 units. Over 150,000 apparel units were demanded with only 3 to 27 units on hand, revealing that static reorder thresholds failed to track actual rolling demand velocity.

Detailed Finding 3: 4,470 Delivery SLA Breaches

Analysis of delivery logs revealed that 4,470 shipments failed the contractual ExpectedDeliveryDate , averaging a 3-day delay. Carrier segmentation proved that delivery failures were concentrated in specific third-party freight providers rather than internal fulfillment delays.

Detailed Finding 4: Overdue Accounts Receivable Concentration (\$267.0K)

84 invoices remained unpaid totaling \$267,011.44. Severe counterparty credit risk was identified: two enterprise buyers—Wingtip Toys (Head Office) (\$97,053.58 across 22 invoices) and Tailspin Toys (Head Office) (\$56,435.84 across 21 invoices)—accounted for 57.5% of total outstanding debt. Zero accounts had been placed on credit hold (IsOnCreditHold = 0).

CAPSTONE PROJECT CASE STUDY

Dynamic RLS Implementation & Dashboard Layout

Production DAX Security Implementation

The following battle-tested DAX expression was applied to `BI.vw_DimCustomer` in Power BI Desktop under the role `TerritoryManagerRole` :

```
VAR CurrentUserPrincipal = USERPRINCIPALNAME()
VAR CurrentLocalUser = USERNAME()

// Retrieve all sales territories assigned to the active user in the security bridge
VAR UserAssignedTerritories =
    CALCULATETABLE(
        VALUES(vw_SecurityUserTerritory[SalesTerritory]),
        vw_SecurityUserTerritory[UserEmail] = CurrentUserPrincipal ||
        vw_SecurityUserTerritory[UserEmail] = CurrentLocalUser
    )

// Check if user has global executive access ('*' wildcard)
VAR HasGlobalExecutiveAccess =
    "*" IN UserAssignedTerritories

RETURN
    HasGlobalExecutiveAccess || vw_DimCustomer[SalesTerritory] IN UserAssignedTerritories
```

Two-Page Operational Dashboard Blueprint

Page 1: Executive Operations & Territory Governance

- **Dynamic Security Header:** Live card displaying active authenticated user UPN, role title, and assigned territory scopes.
- **Executive KPI Strip:** Total Revenue, Operating Profit, Gross Margin % (>50% goal), and alert badges for Unfulfilled Revenue (\$5.37M) and Overdue Debt (\$267K).
- **Sales Trend Visual:** Line & Clustered Column Chart tracking Monthly Revenue vs. Gross Margin % over calendar periods.
- **Territory Geographic Distribution:** Bar Chart displaying regional revenue (under RLS, dynamically filtered to the manager's territory).
- **Enterprise Customer Matrix:** Ranked customer performance with drill-through to invoice lines.

Page 2: Diagnostic RCA & Operations Risk Tracker

- **Negative Margin Scatter Matrix:** Visualizing SKU operating profit against sales volume, isolating Zombie Masks and 10-pack USB pricing errors.
- **Inventory Stockout Depletion Chart:** Clustered Bar Chart comparing unpicked demand units against warehouse stock-on-hand.
- **Aging Receivables Exposure Donut:** Visualizing customer debt concentration, highlighting Wingtip Toys (\$97K) and Tailspin Toys (\$56K).
- **Fulfillment SLA Heatmap:** Carrier delivery delays mapped across shipment priority classes.

CAPSTONE PROJECT CASE STUDY

Benchmarking, Technical Scale & Evaluation

Capstone Artifact Scale & Codebase Metrics

The complete project implementation spans modular SQL scripts, advanced DAX calculation libraries, Power Query M ingestion routines, and comprehensive technical documentation:

Artifact / Code File	Technology	File Size	Lines of Code
SQL_Queries/01_optimized_views_and_queries.sql	T-SQL (DDL/DML)	5.2 KB	174 lines
SQL_Queries/02_root_cause_analysis_queries.sql	T-SQL (Forensics)	4.7 KB	103 lines
SQL_Queries/03_security_and_rls_views.sql	T-SQL (Security)	4.1 KB	88 lines
DAX_Measures/01_sales_performance_measures.dax	DAX Formula	1.5 KB	40 lines
DAX_Measures/02_dynamic_rls_and_dashboard_measures.dax	DAX Security/KPI	3.7 KB	111 lines
PowerQuery_M/01_DirectQuery_Setup.pq	Power Query M	1.2 KB	28 lines
PowerQuery_M/02_NativeQuery_With_Folding.pq	Power Query M	1.9 KB	42 lines
PowerQuery_M/03_Import_StarSchema_Dimensions.pq	Power Query M	1.7 KB	36 lines
Total Capstone Artifacts	SQL + DAX + M + MD	~45 KB	~1,100+ lines

Key Architectural Strengths

Dedicated BI Schema Isolation

Reporting queries are fully decoupled from core transactional OLTP tables via the BI schema, eliminating table-level locking, reducing database blocking, and allowing fine-grained grant/deny permission management.

Enterprise Zero-Trust RLS Governance

Multi-tenant security is enforced dynamically via DAX without maintaining separate reports. Regional managers cannot access unauthorized customer records even if they modify visual filters.

Data-Driven Corrective Action Roadmap

The analysis established clear operational mandates: updating mask list prices to \$28.50 to enforce a 33% gross margin, introducing database constraints (UnitPrice ≥ LastCostPrice), shifting from static presets to dynamic velocity-based inventory replenishment, and automating credit hold flags for accounts overdue beyond 30 days.

Technology Stack & Executive Summary

Technology	Category	Role in Data Analysis Track
Python (3.11+)	Programming Language	Data cleaning, scripting, mathematical modeling, and automated pipelines.
NumPy & Pandas	Data Manipulation	High-performance vectorized operations, aggregation, reshaping, and EDA.
Matplotlib & Seaborn	Statistical Visualization	Distribution plots, correlation heatmaps, regression lines, and outlier charts.
Microsoft Excel 365	Spreadsheet & Analytics	XLOOKUP, dynamic arrays, scenario modeling, PivotTables, and in-sheet ETL.
Microsoft SQL Server	RDBMS & Data Storage	Relational modeling, T-SQL scripting, window functions, indexing, and BI views.
Power BI Desktop	Enterprise BI & Reporting	Star-schema data modeling, visual canvas authoring, and cloud service integration.
DAX (Data Analysis Expressions)	Calculated Formula Language	Measures, evaluation contexts, time intelligence, and dynamic RLS governance.
Power Query (M)	ETL Transformation Engine	Data ingestion, type transformation, column merging, and query folding pushdown.

Final Executive Summary

This training report presents the comprehensive knowledge, methodological rigor, and practical accomplishments achieved throughout the **Data Analysis Track** at **Puresoft**.

Starting with fundamental data manipulation in advanced Microsoft Excel and exploratory programming in Python (NumPy, Pandas, Matplotlib, Seaborn), the curriculum advanced into inferential statistics, relational database design, and high-performance T-SQL query engineering. These foundations culminated in enterprise-grade business intelligence architecture using Microsoft Power BI, star schema modeling, VertiPaq vs. DirectQuery performance profiling, and dynamic multi-tenant Row-Level Security.

The complete skill set was integrated and proven within the **WideWorldImporters Enterprise Capstone Project**. The project successfully diagnosed four systemic operational vulnerabilities: reversing -\$320K in pricing margin inversions, diagnosing \$5.37M in unfulfilled stockout losses, isolating carrier delivery SLA bottlenecks, and mitigating \$267K in overdue accounts receivable concentration. The resulting production solution delivers high-speed, secure, and actionable intelligence to executive decision-makers.

Candidate Signature:

Omar Sharaby

Data Analysis Track — Training Report
Data Analysis Track — Puresoft

Completion Date:

September 2026

Page 16
Verified Training Portfolio